

MULTIMODAL ITERATIVE RAG FOR KNOWLEDGE-INTENSIVE VISUAL QUESTION ANSWERING

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ABSTRACT

Recent advances in Multimodal Large Language Models (MLLMs) have significantly enhanced the ability of these models in multimodal understanding and reasoning. However, the performance of MLLMs for knowledge-intensive visual questions, which require external knowledge beyond the visual content of an image, still remains limited. While Retrieval-Augmented Generation (RAG) has become a promising solution to provide models with external knowledge, its conventional single-pass framework often fails to gather sufficient knowledge. To overcome this limitation, we propose MI-RAG, a Multimodal Iterative RAG framework that leverages reasoning to enhance retrieval and incorporates knowledge synthesis to refine its understanding. At each iteration, the model formulates a reasoning-guided multi-query to explore multiple facets of knowledge. Subsequently, these queries drive a joint search across heterogeneous knowledge bases, retrieving diverse knowledge. This retrieved knowledge is then synthesized to enrich the reasoning record, progressively deepening the model’s understanding. Experiments on challenging benchmarks, including Encyclopedic VQA, InfoSeek, and OK-VQA, show that MI-RAG significantly improves both retrieval recall and answer accuracy, establishing a scalable approach for compositional reasoning in knowledge-intensive VQA.

1 INTRODUCTION

The emergence of Multimodal Large Language Models has driven significant progress in multimodal understanding and reasoning. Nonetheless, the performance of MLLMs remains constrained in knowledge-intensive Visual Question Answering (VQA) tasks, which require external knowledge beyond the visual content in the image. Answering such questions is particularly challenging as it involves: (1) grounding visual entities in the image to the textual question, (2) collecting text knowledge relevant to both the entities and the question, and (3) synthesizing the visual entities and corresponding textual knowledge to derive the answer.

To address this challenge, multimodal Retrieval-Augmented Generation has emerged as a promising solution. The conventional *retrieve-then-read* process starts with a retriever using the input image and question to find relevant image-text pairs from a knowledge base (KB). A reader model then synthesizes this information to create the final answer. Research on optimizing this single-step process has largely advanced along two fronts: improving initial retrieval with high-quality embeddings Lin et al. (2024); Wei et al. (2024); Jiang et al. (2025); Lin et al. (2025); Jiang et al. (2024b); Zhang et al. (2024b), and refining retrieval results by selection and reranking Cocchi et al. (2025); Zhang et al. (2024a); Ling et al. (2025); Chen et al. (2024); Liu et al. (2024b); Yan & Xie (2024); Yang et al. (2025).

Although prior work has achieved notable improvements within the *retrieve-then-read* process, the process itself has inherent limitations for knowledge-intensive VQA. A single retrieval step with an initial query is often insufficient, as imperfect retrievers may fail to gather all necessary knowledge at once Shi et al. (2023); Zhang et al. (2023); Cuconasu et al. (2024). Furthermore, a single reasoning step to ground visual entities and synthesize knowledge is problematic, as distracting or irrelevant

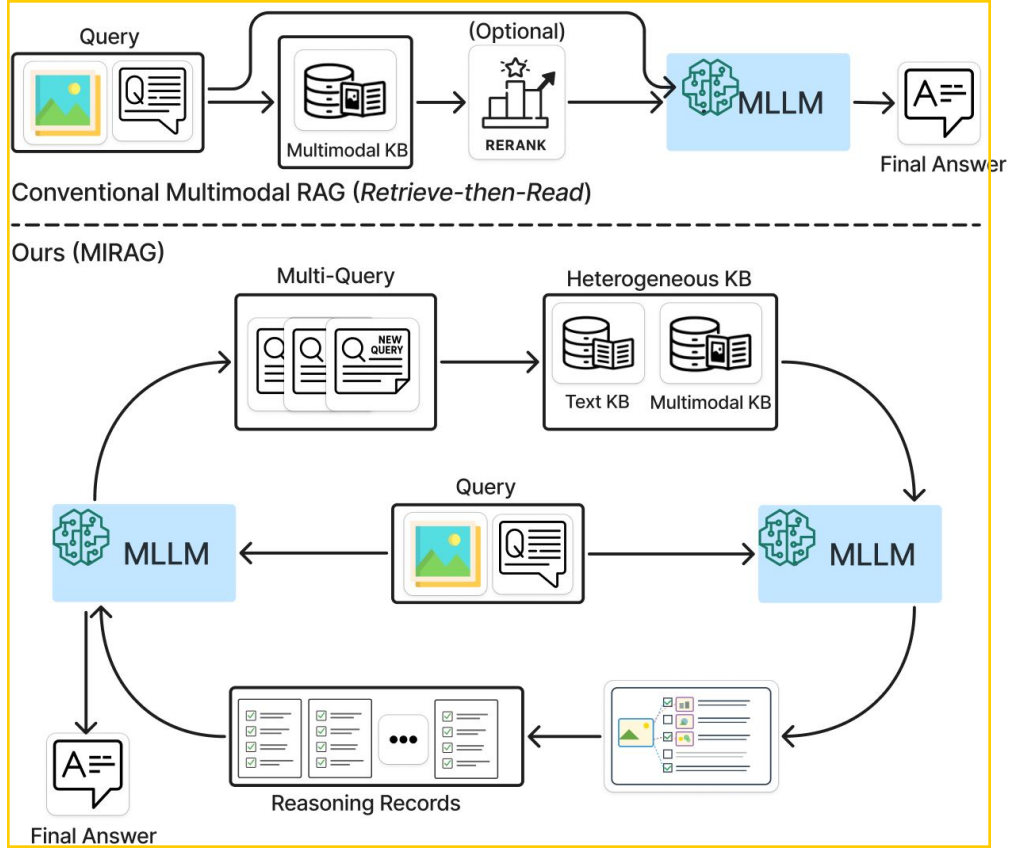


Figure 1: An overview of conventional multimodal RAG and our MI-RAG framework. Conventional multimodal RAG employs a *retrieve-then-read* process using only a multimodal KB. In contrast, MI-RAG utilizes an iterative framework that cyclically refines its query and reasoning. At each iteration, MI-RAG formulates a multi-query with accumulated reasoning records and retrieves diverse knowledge from heterogeneous KBs. It then performs compositional reasoning to synthesize newly acquired knowledge into a reasoning record and appends it to the reasoning records to improve the subsequent iteration.

content can mislead synthesis and influence the final answer Petroni et al. (2020); Wu et al. (2024); Amiraz et al. (2025); Huang et al. (2025); Wu et al. (2025).

Motivated by these limitations, the text domain has shifted to *iterative RAG*, a paradigm that leverages an LLM’s reasoning through multiple rounds of retrieval and reasoning. In this iterative cycle, the model leverages its reasoning to guide intermediate steps, which transform queries to collect more relevant knowledge and synthesize newly retrieved knowledge to progressively update the reasoning itself. Shao et al. (2023); Trivedi et al. (2022); Xiong et al. (2024a); Liu et al. (2024a); Yue et al. (2025); Gao et al. (2025). While previous work has shown promise with this approach in the text-only domain, it has largely focused on refining a single subquery of decomposing multi-hop questions at each step. Extending this to the more complex multimodal domain remains an open challenge.

This challenge arises because a naive adaptation of the *iterative RAG* is insufficient for the multimodal domain, which introduces two distinct considerations: (i) it requires retrieving diverse knowledge by exploring multiple facets of a visually grounded entity and its related textual knowledge across modalities, and (ii) it demands compositional reasoning to synthesize the visual entity with its corresponding knowledge by integrating a diverse set of visual-to-visual, visual-to-text, and text-

to-text factual links. This highlights the need for a more specialized approach that can effectively handle the complexities of both visual and textual knowledge.

In response to these multimodal domain considerations, we propose Multimodal Iterative Retrieval-Augmented Generation (MI-RAG). As illustrated in Figure 1, MI-RAG is an iterative framework that cyclically refines its query and reasoning by alternately performing reasoning-guided query transformation and retrieval-augmented reasoning. Our framework makes two primary contributions. First, we employ a reasoning-guided multi-query that dynamically searches multiple facets across modalities. Second, these queries drive a joint search across heterogeneous KBs to find a diverse set of factual links, allowing MI-RAG to compose visually-grounded knowledge and related textual knowledge. Through the synergy of multi-query and heterogeneous KBs, MI-RAG is able to retrieve sufficient knowledge and enable robust compositional reasoning over diverse knowledge across modalities. Through iterations, MI-RAG collects diverse knowledge progressively with accumulating a reasoning record, where each record is the result of compositional reasoning over newly retrieved knowledge, thereby refining the model’s understanding to guide the subsequent iteration step.

2 RELATED WORK

2.1 MULTIMODAL RAG FOR KNOWLEDGE-INTENSIVE VQA

The dominant paradigm for multimodal RAG follows the *retrieve-then-read* process. Initial work focused on enhancing the retrieval phase, primarily by learning more effective multimodal embeddings. Pioneering approaches aligned queries and text documents using contrastive learning Luo et al. (2023), while others fused similarity scores from pretrained CLIP Wei et al. (2024). To improve retrieval, subsequent methods employed fine-grained late interaction between visual and textual tokens Lin et al. (2023; 2024) or leveraged MLLMs with instruction tuning to produce multimodal embeddings Jiang et al. (2025); Liu et al. (2024b); Zhang et al. (2024b). A separate line of research has explored generative retrieval, using document identifiers Long et al. (2024) or enhancing generative retrieval via reinforcement learning Long et al. (2025).

Another line of research improves the read phase, typically by reranking or refining the retrieved candidates. Hierarchical systems first perform a coarse, image-based search and then refine the results with text-based or multimodal rerankers Caffagni et al. (2024); Yan & Xie (2024); Yang et al. (2025). Others leverage the reasoning capabilities of MLLMs to self-reflect on retrieval needs and evaluate relevance Cocchi et al. (2025), generate semantic tags for consistency and reranking based on relevance scores Ling et al. (2025). While effective, these methods largely adhere to the *retrieve-then-read* process. In contrast, MI-RAG introduces a dynamic loop that iterates between reasoning-guided query refinement and retrieval-augmented reasoning to progressively solve knowledge-intensive VQA tasks.

2.2 ITERATIVE RAG

When single-step retrieval is insufficient for complex, multi-hop questions, *iterative RAG* has emerged as a powerful paradigm in the text-only domain. These methods typically use an iterative loop to refine queries and retrieve new knowledge to improve subsequent iterations. Query transformation techniques are central to this process and include query expansion, which uses pseudo-answers to formulate richer queries Shao et al. (2023); Wang et al. (2024); query generation, which decomposes complex questions into simpler, answerable sub-queries Zhang et al. (2025); Liu et al. (2024a); and query planning, where a model strategically decides when and what to retrieve Trivedi et al. (2022); Yu et al. (2024).

To guide the subsequent query transformation, the LLM performs retrieval-augmented reasoning. This involves synthesizing newly retrieved information into a reasoning record. Common strategies include summarizing retrieved passages Jiang et al. (2024c); Zhang et al. (2025), extracting direct evidence snippets Yu et al. (2024), or generating pseudo-answers for intermediate sub-queries, often with chain-of-thought prompting Shao et al. (2023); Trivedi et al. (2022). However, these text-based iterative methods are primarily designed for multi-hop question decomposition, where the goal is to build a linear reasoning chain. Our work addresses knowledge-intensive VQA, which requires the composition of diverse knowledge across modalities.

3 METHOD

3.1 FRAMEWORK OF MULTIMODAL ITERATIVE RAG

Algorithm 1 MI-RAG

Require: $I, Q, \text{KB}_T, \text{KB}_M, \text{MLLM}, N$
Ensure: Final answer A

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1:  $D \leftarrow \text{MLLM}(Q, I)$ 
2:  $Q_0^E \leftarrow \text{Expand}(Q, D)$ 
3:  $K_T \leftarrow \text{Ret}(\text{KB}_T, Q_0^E)$ 
4:  $K_M \leftarrow \text{Ret}(\text{KB}_M, I, Q_0^E)$ 
5:  $R_0 \leftarrow \text{Reasoning}(\text{MLLM}, Q, I, K_T, K_M, D)$ 
6: for  $i = 1$  to  $N$  do
7:    $Q_i^E \leftarrow \text{Expand}(Q, R_{i-1})$ 
8:    $Q_i^G \leftarrow \text{Generate}(\text{MLLM}, Q, R_0, \dots, R_{i-1})$ 
9:    $K_T \leftarrow \text{Ret}(\text{KB}_T, Q_i^E)$ 
10:   $K_T \leftarrow K_T \cup \text{Ret}(\text{KB}_T, Q_i^G)$ 
11:   $K_M \leftarrow \text{Ret}(\text{KB}_M, I, Q_i^E)$ 
12:   $K_M \leftarrow K_M \cup \text{Ret}(\text{KB}_M, I, Q_i^G)$ 
13:   $R_i \leftarrow \text{Reasoning}(\text{MLLM}, Q, I, K_{\text{text}}, K_{\text{mm}})$ 
14: end for
15:  $A \leftarrow \text{MLLM}(Q, I, \{R_0, R_1, \dots, R_N\})$ 
16: return  $A$ 

```

knowledge about the queries (Line 13; Section 3.5). Finally, the model derives the final answer over the accumulated records (Line 15; Section 3.6).

As detailed in Algorithm 1, the MI-RAG framework processes multimodal queries (an image I and question Q), through iterative retrieval and reasoning. For conciseness in the pseudocode, we use the following abbreviations: KB_T for text KB, KB_M for multimodal KB, K_T for text passages, K_M for image-text pairs, Ret for Retrieve, D for image description, Q^E for expanded query, Q^G for generated sub-query, R for reasoning record and N for iteration counts. This iterative process begins with the generation of initial reasoning records (Lines 1–5; Section 3.2). Subsequently, in each iteration, MI-RAG leverages the accumulated reasoning record to formulate the multi-query (Lines 7–8; Section 3.3) and perform a joint search using the multi-query across heterogeneous KBs (Lines 9–12; Section 3.4), thereby dynamically gathering diverse and relevant knowledge. Following retrieval, MI-RAG updates the reasoning record by summarizing newly acquired

3.2 INITIAL REASONING RECORD GENERATION

The initial reasoning record is generated by leveraging the retrieved knowledge with internal MLLM knowledge. Specifically, given a multimodal query, the MLLM generates an image description D corresponding to the question. We perform query expansion, yielding expanded query $Q_0^E = [Q, D]$. This query Q_0^E is then used to retrieve knowledge from our heterogeneous KBs, collecting both text passages and image-text pairs. Finally, the model synthesizes retrieved diverse knowledge and descriptions into a reasoning record. This initial process generates the first reasoning record to bootstrap our iterative framework.

3.3 REASONING-GUIDED MULTI-QUERY TRANSFORMATION

MI-RAG employs a reasoning-guided multi-query transformation that leverages the MLLM’s intermediate reasoning steps to dynamically reformulate the query. This process uses the accumulated reasoning records to dynamically generate a set of complementary queries, ensuring a comprehensive search for multifaceted knowledge across modalities:

Query Expansion: This method leverages the latest reasoning record to expand the original query. The resulting query is designed to reflect the model’s latest reasoning, which enhances retrieval performance while preserving the original question’s intent.

Query Generation: This method produces a set of sub-queries designed to explore alternative reasoning paths, resolve ambiguities, or explicitly seek out missing facts identified during the analysis of the reasoning record.

We provide details of the prompt templates used to guide this transformation in Appendix C.

3.4 JOINT SEARCH ACROSS HETEROGENEOUS KBs

To enhance compositional reasoning, it is crucial to retrieve diverse knowledge across modalities. To acquire the a diverse set of factual knowledge, we perform a joint search over two heterogeneous KBs: a multimodal KB and a textual KB. The multimodal KB provides image-text pairs for visual

grounding of entities, while the textual KB offers broad knowledge that extends beyond these pairs. Each query from the multi-query is used to retrieve candidates from both KBs simultaneously. To balance the token budget with broad knowledge coverage, our strategy retrieves twice as many text passages as image-text pairs, collecting up to 20 passages and 10 image-text pairs per iteration. The implementation details for multimodal retrieval are provided in Appendix D.

Retrieval from textual KB: The textual KB is queried using the multi-query. A text embedding model is used to retrieve the top-k passages based on semantic similarity.

Retrieval from multimodal KB: In contrast to prior methods that often rely on large, specialized retrievers, MI-RAG employs a lightweight retrieval strategy. We utilize a multimodal similarity scoring approach Changin et al. (2025) to support query transformation. For each candidate image-text pair, the similarity score is the average of two components: (1) a text-to-text similarity between the candidate’s text and the transformed query; and (2) a fixed image-to-image similarity between the candidate’s image and the input query image, which maintains constant visual relevance across iterations.

3.5 GENERATE REASONING RECORD WITH KNOWLEDGE

At each iteration, a reasoning record is generated by summarizing the newly retrieved knowledge. The MLLM synthesizes relevant evidence from visual entities and related text knowledge. This process utilizes the MLLM’s compositional reasoning to generate a reasoning record that captures relevant evidence. By iteratively enriching its understanding with these reasoning records, the model maintains a coherent reasoning path and enables more precise query transformations in subsequent steps. We provide details of the prompt in Appendix C.

3.6 FINAL ANSWER GENERATION

Upon completion of the N iterative cycles, MI-RAG generates the final answer. In this stage, MLLM synthesizes the original question Q , the input image I , and the accumulated reasoning records $\{R_0, \dots, R_N\}$ to produce an evidence-based answer.

4 EXPERIMENTS

4.1 EXPERIMENT SETUP

Datasets: Our evaluation is conducted on three diverse knowledge-intensive VQA benchmarks: validation split of InfoSeek Chen et al. (2023), validation split of OK-VQA Marino et al. (2019), and test split of Encyclopedic VQA Mensink et al. (2023). For InfoSeek, we follow the setup of PreFLMR Lin et al. (2024) and utilize the validation split provided by the M2KR benchmark.

Knowledge Bases: To provide the necessary external knowledge, our framework retrieves text passages and image-text pairs from heterogeneous KBs, which are held constant across all experiments: a Wikipedia text corpus provided by FlashRAG Jin et al. (2024) for text KB and a 2M image-text pairs from Encyclopedic VQA for multimodal KB.

Models: To assess how performance scales with reasoning ability, we primarily use two model variants: Gemma-3 Team et al. (2025) and Gemini-2.5-Flash Comanici et al. (2025). Our framework employs lightweight retrievers. For the multimodal retrieval of image-text pairs, we compute a multimodal similarity score using image embeddings from SigLIP2-SO-400m Tschannen et al. (2025) and text embeddings from ModernBERT Warner et al. (2024). For the retrieval of text passages, we use E5-base-v2 Wang et al. (2022), following the setup in FlashRAG Jin et al. (2024).

Evaluation Metrics: We evaluate MI-RAG using standard metrics for each benchmark to evaluate accuracy and retrieval recall. For accuracy, we report the BERT matching score (BEM) Bulian et al. (2022) on Encyclopedic VQA. On the InfoSeek dataset, we measure Exact Match (EM) accuracy. To constrain the model’s answer to a single exact entity for this metric, we employ a 4-shot prompting strategy using examples from the training split. The full prompt template and further details are available in Appendix E. For OK-VQA, we use the standard VQA-score Marino et al. (2019) and a Cover Exact Match (Cover EM) Jiang et al. (2024a); Yue et al. (2025), which checks if the ground-

truth answer is contained within the response. To assess retrieval performance, we use two metrics based on ground-truth annotation availability. Retrieval quality is measured using Recall of ground-truth entities where available, and Pseudo-Relevance Recall (PRR) otherwise Luo et al. (2021).

4.2 RETRIEVAL PERFORMANCE ON VQA BENCHMARKS

Table 1: Recall comparison on the InfoSeek and Encyclopedic VQA. The FT column with $\checkmark$ indicates methods fine-tuned on the target dataset. Methods marked with $\ddagger$ employ a reranking step after retrieval. Our vanilla method (i) uses a single query with expansion to retrieve K pairs from the multimodal KB, reporting average recall across iterations for fair comparison. Our full framework (*) reports the cumulative recall after 4 iterations. We report it on $R@5$, as each query retrieves fewer than five pairs on average. The best and second-best overall accuracies are highlighted in **bold** and with an underline, respectively.

FT	Method	InfoSeek Validation			Encyclopedic VQA		
		R@5	R@10	R@20	R@5	R@10	R@20
$\times$	CLIP ViT-L/14 Radford et al. (2021)	54.0	61.6	68.6	7.7	12.1	16.5
$\times$	SigLIP2-So400m Tschannen et al. (2025)	52.5	60.2	68.3	30.8	36.6	41.9
$\times$	EVA-CLIP-8B Sun et al. (2023)	67.1	73.0	77.9	31.3	41.0	48.8
$\checkmark$	Wiki-LLaVA Caffagni et al. (2024)	-	66.1	71.9	-	9.9	13.2
$\checkmark$	LLM-RA Jian et al. (2024)	53.8	-	-	-	-	-
$\checkmark$	mR ² AG $\ddagger$ Zhang et al. (2024a)	-	65.0	71.0	-	-	-
$\checkmark$	ReflectiVA $\ddagger$ Cocchi et al. (2025)	<u>77.6</u>	-	<u>86.4</u>	36.1	-	<u>49.8</u>
$\checkmark$	EchoSight $\ddagger$ Yan & Xie (2024)	74.0	77.4	77.9	<u>47.9</u>	<u>48.8</u>	48.8
$\checkmark$	OMGM Yang et al. (2025)	73.9	80.0	84.8	41.2	49.8	58.7
$\checkmark$	OMGM $\ddagger$ Yang et al. (2025)	80.8	<u>83.6</u>	84.8	55.7	58.1	58.7
$\times$	ReAuSE Long et al. (2025)	59.5	-	-	-	-	-
$\times$	Ours (Gemma-3-4B) $\ddagger$	80.8	85.6	89.7	39.1	44.3	<u>49.8</u>
$\times$	Ours (Gemma-3-4B)*	88.0	-	-	53.6	-	-
$\times$	Ours (Gemini-Flash-2.5)*	92.1	-	-	62.4	-	-

As shown in Table 1, our MI-RAG framework demonstrates effective retrieval without any task-specific fine-tuning. We analyze its performance by highlighting three key findings: the effectiveness of reasoning-guided queries in a lightweight setup, MI-RAG’s performance boost from the iterative process, and the framework’s inherent synergy with larger MLLMs.

To evaluate our reasoning-guided query transformation, we provide a vanilla method (i), which uses a single query with a multimodal KB, a small 4B MLLM, and a lightweight retriever for fair comparison. This contrasts with approaches that employ larger retrievers and use MLLM for reranking. Even under these constraints, our method is highly effective, achieving a top $R@20$ score of 89.7% on InfoSeek and the second-highest $R@20$ on Encyclopedic VQA.

Our full framework (*) performs multiple rounds of multi-query searches across heterogeneous KBs to progressively accumulate multifaceted knowledge across modalities. The results show significant gains in cumulative recall, reaching 88.0% (+7.2%) on InfoSeek and 53.6% (+14.5%) on Encyclopedic VQA.

Finally, we analyze MI-RAG’s scalability by evaluating its two core components: the MLLM and the retriever. As shown in Table 1, scaling the MLLM from Gemma-3-4B to Gemini-Flash-2.5 consistently improves performance. This result confirms that a more capable MLLM enhances both query transformation and compositional reasoning. In Section 6.2, we demonstrate that scaling the retriever provides a further performance boost.

On the OK-VQA benchmark, MI-RAG demonstrates strong retrieval performance, as detailed in Table 2. Our vanilla method (i), despite its lightweight setup, shows competitive performance. Enabling the full MI-RAG provides a substantial boost to 95.2% (18.1%) for the same 4B model. A more powerful MLLM with stronger reasoning achieves 97.1% recall, outperforming all prior work.

This robust performance contrasts sharply with prior methods, which often exhibit a performance trade-off across different benchmarks. For instance, ReAuSE excels in its trained knowledge do-

Table 3: Accuracy comparison on the Encyclopedic VQA benchmark. A checkmark (✓) indicates fine-tuning, a dagger (†) denotes MLLM-based reranking, and an asterisk (*) marks the use of ReflectiVA checkpoints. Our framework’s Gemma-series models generate their own reasoning records, whereas other models use records provided by Gemini-2.5-Flash.

Google Lens	FT	Method	Model	Single-hop	All
×	✓	Wiki-LLaVA Caffagni et al. (2024)	LLaVA-v1.5-7B	21.8	–
	✓	EchoSight† Yan & Xie (2024)	Mistral-7B	35.5	–
	✓	ReflectiVA† Cocchi et al. (2025)	LLaVA-MORE-8B	35.5	–
	✓	MMKB-RAG† Ling et al. (2025)	Qwen2-VL-7B	39.7	35.9
	✓	OMGM† Yang et al. (2025)	LLaVA-v1.5-7B	50.2	–
	✓	OMGM† Yang et al. (2025)	GPT-4o	51.2	–
	✓	Ours	LLaVA-MORE-8B*	50.4	49.2
	×	Ours	Gemma-3-4B	48.4	52.4
✓	×	Ours	Gemini-2.5-Flash	61.4	62.9
	×	HAMMR Castrejon et al. (2024)	PaLI-X	47.8	–
	×	LVLm cascade Alazraki et al. (2023)	PaLM, GPT-3, PaLI	53.4	–
	✓	mR ² AG† Zhang et al. (2024a)	LLaVA-v1.5-7B	55.9	51.8
	×	Ours	Gemma-3-4B	62.8	63.7
	×	Ours	Gemini-2.5-Flash	75.5	70.8

main (OK-VQA) but fails to generalize to InfoSeek, whereas OMGM shows the reverse pattern. In contrast, our method shows high performance across all benchmarks.

4.3 ACCURACY ON VQA BENCHMARKS

Table 2: Recall comparison on the OK-VQA benchmark when utilizing Wikipedia as the knowledge source. Symbols denote: fine-tuning (✓); reranking (†); our vanilla baseline (‡); and our MI-RAG framework (*). The best and second-best overall accuracies are highlighted in **bold** and with an underline, respectively.

FT	Methods	OK-VQA		
		PRR@5	PRR@10	PRR@20
✓	ReViz-ICT	61.9	72.6	<u>81.1</u>
✓	GeMKR	70.8	79.1	–
✓	FLMR	68.1	78.0	–
✓	Pre-FLMR	68.6	–	–
✓	ReAuSE	88.0	91.3	–
×	OMGM†	73.4	–	–
×	Ours (Gemma-3-4B)‡	<u>77.1</u>	<u>85.4</u>	91.2
×	Ours (Gemma-3-4B)*	95.2	–	–
×	Ours (Gemini-Flash-2.5)*	97.1	–	–

As demonstrated in Table 3, our MI-RAG framework establishes a new state-of-the-art on the Encyclopedic VQA benchmark. To isolate the contribution of our iterative reasoning, we first evaluate a setting with a strong initial retrieval provided by Google Lens. This configuration reveals that performance scales directly with the MLLM’s reasoning capacity: accuracy improves from a competitive 62.8% with Gemma-3-4B to a state-of-the-art 75.5% with Gemini-2.5-Flash. This highlights the synergy between a high-quality initial retrieval and the model’s ability to iteratively refine it, an analysis further explored in Section 6.2.

In the more challenging setting without Google Lens, we use reasoning records generated by Gemini-2.5-Flash with our lightweight retriever

setup. When these records are applied to LLaVA-MORE-8B checkpoint from ReflectiVA as the answerer, our framework achieves a competitive accuracy of 50.4%. When Gemini-2.5-Flash is used as the answerer, our method achieves 61.4% accuracy.

On the InfoSeek benchmark, MI-RAG translates its strong retrieval performance into state-of-the-art accuracy, as detailed in Table 4. Our method achieves 48.0% accuracy with Gemini-2.5-Flash and improves to 51.7% with GPT-4o as the answerer, outperforming all prior work. This is driven by our iterative process, where accumulated reasoning records enable robust compositional reasoning over multifaceted, diverse knowledge. The strength of this iterative reasoning is further demonstrated by its ability to improve ReflectiVA’s LLaVA-MORE-8B checkpoint to 45.1%.

Recent MLLMs generate detailed, free-form responses by leveraging multi-stage post-training to better align with human preferences, making strict exact match (EM) an inadequate metric for a comprehensive evaluation, as noted in prior work Shao et al. (2023); Cocchi et al. (2025). Conse-

Table 4: Accuracy comparison on the InfoSeek benchmark: the top section evaluates the full validation set, while the bottom section assesses a 5K validation split. A checkmark (✓) in the FT column indicates fine-tuning, a dagger (†) denotes a reranking step, and an asterisk (*) marks the use of checkpoints from ReflectiVA. Our framework’s Gemma-series models generate their own reasoning records, whereas other models use records provided by Gemini-2.5-Flash.

FT	Method	Retriever	Model	Acc
✓	Wiki-LLaVA Caffagni et al. (2024)	CLIP-ViT-L	LLaVA-1.5-7B	28.9
✓	EchoSight† Yan & Xie (2024)	EVA-CLIP-8B	LLaMA3-8B	31.3
✓	mR ² AG† Zhang et al. (2024a)	CLIP-ViT-L	LLaVA-1.5-7B	40.2
✓	ReflectiVA† Cocchi et al. (2025)	EVA-CLIP-8B	LLaVA-MORE-8B	40.1
✓	MMKB-RAG† Ling et al. (2025)	PreFLMR ViT-G	Qwen2-VL-7B	36.7
✓	OMGM† Yang et al. (2025)	EVA-CLIP-8B	LLaVA-1.5-7B	43.5
✓	OMGM† Yang et al. (2025)	EVA-CLIP-8B	GPT-4o	42.1
✓	RA-VQA-v2 Lin et al. (2024)	PreFLMR ViT-G	BLIP2-Flan-T5-XL	30.7
✓	MMKB-RAG Ling et al. (2025)	PreFLMR ViT-G	Qwen2-VL-7B	34.7
✓	MUKA Deng et al. (2025)	MUKA	VILA-13B	42.5
✓			LLaVA-MORE-8B*	45.1
×			Gemma-3-4B	33.1
×	Ours	SigLIP2-SO-400m	Gemini-2.5-Flash	48.0
×			GPT-4o	51.7

quently, metrics like Cover EM (CEM), which verify the inclusion of the ground-truth answer within a longer response, are increasingly used Jiang et al. (2024a); Yue et al. (2025); Wang et al. (2025). This gap arises when MLLM is not explicitly fine-tuned to produce a strict single-entity answer. The MLLM’s tendency to generate a rationale before the final answer results in penalties under the EM metric.

5 ABLATIONS

Table 5: Comparison of accuracy based on two metrics: Cover EM, which checks if the ground-truth answer is included in the response, and Exact Match, which requires the response to match the ground-truth answer exactly.

Model	InfoSeek Validation		InfoSeek Human		OK-VQA	
	EM	CEM	EM	CEM	EM	CEM
Gemma-3-4B	33.1	50.4	32.3	39.3	43.3	68.8
Gemini-2.5-Flash	48.0	61.3	40.5	44.9	57.3	71.0

across both single-query and multi-query. We report performance using cumulative recall, along with CoverEM on InfoSeek and BEM on Encyclopedic VQA. We provide details of the downsampling algorithm in Appendix G.

We conducted ablation studies to evaluate MIRAG’s core components. For efficiency, all experiments are performed on downsampled subsets of InfoSeek and Encyclopedic VQA using Gemini-2.5-Flash as the MLLM. We strictly controlled the retrieval budget to ensure a fair comparison. For our default heterogeneous KB setting, we retrieve 20 text passages and 10 image-text pairs. To maintain a comparable budget for multimodal KB, we retrieve 20 pairs. This total retrieval budget is held constant

5.1 ABLATION OF COMPONENTS

Table 6: Ablation of MIRAG on the InfoSeek validation subset and Encyclopedic VQA test subset.

Components			InfoSeek		Encyclopedic VQA	
Multi Query	Hetero. KB	Iter.	Acc	Recall	Acc	Recall
×	×	×	46.70	87.28	50.78	47.06
✓	×	×	49.44	90.34	52.94	49.02
×	✓	×	49.21	88.08	51.18	52.75
✓	✓	×	54.11	91.43	56.67	57.25
✓	✓	✓	61.84	94.36	63.33	67.25

Our ablation analysis in Table 6 reveals the impact of each component. We establish a strong baseline using a vanilla RAG system over a multimodal KB, enhanced with our initial query expansion, which integrates the MLLM’s knowledge. Introducing the reasoning-guided multi-query yields a substantial accuracy gain by gathering the multifaceted knowledge. Moreover, employing our heterogeneous KB is crucial for compositional reasoning, as it en-

ables the model to compose visually-grounded knowledge with the broad textual knowledge. Combining these two components creates a crucial synergy for compositional reasoning. However, the iterative process delivers the most significant improvement across all datasets.

5.2 IMPACT OF ITERATIVE REFINEMENT

Table 7: Performance over iterations on the InfoSeek and Encyclopedic VQA.

Iter.	InfoSeek		Encyclopedic VQA	
	Acc	Recall	Acc	Recall
1	55.72	92.13	58.46	65.10
2	59.10	92.95	60.86	66.27
3	61.41	93.44	61.96	66.86
4	61.84	94.36	63.33	67.25

To further investigate the benefits of iterative refinement, we analyze performance across multiple iterations, with results presented in Table 7. The study confirms that iterative processing is a primary driver of accuracy, and performance consistently improves over the vanilla baseline across all datasets. The most substantial gains are typically observed within the first two iterations, validating the efficacy of the core refinement loop. As visualized in Figure 2, these gains gradually saturate.

5.3 IMPACT OF HETEROGENEOUS KB

Table 8: Ablation on the KB configuration, showing performance after 4 iterations. Heterogeneous KB refers to the use of both KBs, while Multimodal restricts retrieval to the multimodal KB only.

KB Config.	InfoSeek		Encyclopedic VQA	
	Acc	Recall	Acc	Recall
Multimodal	57.49	91.79	56.86	58.63
Heterogenous	61.84	94.36	63.33	67.25

As shown in Table 8, we evaluate the impact of heterogeneous KB. Applying our reasoning-guided multi-query approach to the multimodal KB alone yields high recall, demonstrating its ability to retrieve visually grounded information. However, its precision remains limited, indicating that relying solely on visually grounded knowledge is often insufficient for compositional reasoning. By combining heterogeneous sources, MI-RAG enables the model to perform compositional reasoning more effectively, leveraging diverse and relevant evidence to refine its search and generate more accurate reasoning records for the final answer.

6 ANALYSIS

6.1 ANALYSIS OF ITERATIVE PERFORMANCE

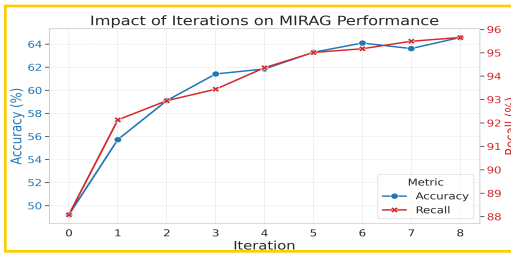


Figure 2: Accuracy and recall of MI-RAG on InfoSeek subset across 9 iterations.

We analyze how iterative refinement impacts MI-RAG’s performance by measuring accuracy and recall across iterations. As shown in Figure 2, performance improves consistently with each step. The initial iterations deliver significant gains. Although the rate of improvement moderates in later steps, the model continues to achieve substantial gains in accuracy and recall. This performance improvement comes with an associated computational cost for each iteration, which we detail in Appendix F.

6.2 SCALING RETRIEVERS

Building upon the established benefits of scaling MLLMs, we now investigate the impact of scaling the retriever. We investigated this by employing a larger vision encoder (SigLIP2-g) and a text embedding model (Qwen3-Embedding-0.6B). On the InfoSeek validation split, when using Gemini-2.5-Flash for MI-RAG, this enhanced retrieval improves accuracy to 66.7% (+6.4%) and cumulative recall to 95.2% (+3.1%). Similarly, on the Encyclopedic VQA test split, the scaled retriever paired with the Gemma-3-4B enhances accuracy to 50.2% (+1.9%) and cumulative recall to 65.7%

(+12.1%). These findings confirm that improving retrieval capability provides significant benefits for MI-RAG’s performance, validating the synergistic design.

6.3 TOP-K SENSITIVITY ANALYSIS

Table 9: Top-k sensitivity analysis on the InfoSeek validation subset. k indicates the number of retrieved image-text pairs.

Model	Metric	$k = 10$	$k = 20$
Gemma-3-4B	Acc	52.66	48.95
	Recall	90.82	94.52
Gemini-2.5-Flash	Acc	61.84	65.70
	Recall	94.36	95.97

Our sensitivity analysis on the InfoSeek validation subset (Table 9) reveals that the benefit of a larger retrieval budget is only beneficial if the MLLM is capable enough to handle the additional context. As shown in Table 9, the results reveal that simply increasing the retrieval budget is not always beneficial. For instance, Gemma-3-4B shows a degradation in accuracy when k is increased from 10 to 20, suggesting that the additional context may potentially distract, hindering its ability to gather the correct ev-

idence.

7 CONCLUSION

In this work, we introduce MI-RAG, a novel multimodal iterative RAG framework that addresses key challenges in knowledge-intensive VQA. MI-RAG employs a reasoning-guided multi-query transformation that dynamically explores multiple facets of an entity and its related textual knowledge across modalities. To facilitate compositional reasoning, its iterative process uses this retrieved evidence to progressively synthesize visual entities and corresponding textual knowledge, effectively integrating a diverse set of factual links. Our results demonstrate that MI-RAG is an effective and scalable paradigm for advancing compositional reasoning in knowledge-intensive VQA. Furthermore, our analysis indicates clear pathways for future work, such as scaling components by integrating web-scale search or dynamically adjusting the retrieval budget based on model capability.

8 ETHICS STATEMENT

Our work introduces new methods for multimodal retrieval-augmented generation for knowledge-intensive visual question answering, and we acknowledge the associated ethical implications. Potential risks include the propagation of biases from the underlying knowledge bases, which can lead to the generation of inaccurate or misleading information. To mitigate these concerns, our framework is intended for research purposes and requires careful evaluation before any deployment. Our approach uses only publicly available datasets and knowledge bases and does not involve private or personally identifiable information. We advocate for continued research on methods for bias detection and mitigation to ensure responsible development of these technologies.

9 REPRODUCIBILITY STATEMENT

To ensure full reproducibility, we provide the main algorithm code for MI-RAG in the supplementary materials. This includes comprehensive scripts for preparing the evaluation data, constructing the knowledge bases, and executing the main experiments. All implementation details, including our core algorithm, are described in Section 3 and detailed in Algorithm 1. The experimental setup, including datasets, knowledge bases, model variants, and evaluation metrics, is thoroughly outlined in Section 4.1.

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A QUALITATIVE EXAMPLES

This section presents a qualitative case study illustrating how MI-RAG constructs reasoning records at each iteration and arrives at the final answer. All examples are drawn from the InfoSeek validation set using MI-RAG with Gemini-2.5-Flash. Figures A1–A10 adopt the same layout described below.

- **Upper-left panel:** The text question and the input image. A descriptive caption about its entity is shown to the right of the image for demonstration only.
- **Lower-left panel:** This panel shows MI-RAG’s final prediction, the ground-truth answer, and the evaluation result. Correct predictions are highlighted in green, while incorrect predictions are in red.
- **Right panel:** This panel presents a four-iteration summary of the MI-RAG reasoning process. The color of each step indicates its reasoning quality with respect to the available evidence: green signifies correct reasoning, yellow is for partially valid or inconclusive steps, red marks an error, and blue is used for emphasis only. Furthermore, any steps following an incorrect one are shaded gray to show they are built upon a previous error.

A.1 SUCCESSFUL CASE ANALYSIS

We present several successful cases that demonstrate the strengths of our MI-RAG framework. These examples illustrate how the iterative process of reasoning-guided retrieval and retrieval-augmented reasoning enables the model to correctly answer knowledge-intensive visual questions.

Successful Visual Identification. A common success pattern begins with accurate recognition of visual entities, followed by a search for related cultural and historical knowledge. In Figure A1, the entity is first identified as a submarine sandwich. MI-RAG then attributes its cultural origins to Italian-American communities and identifies its historical emergence in the Northeastern United States.

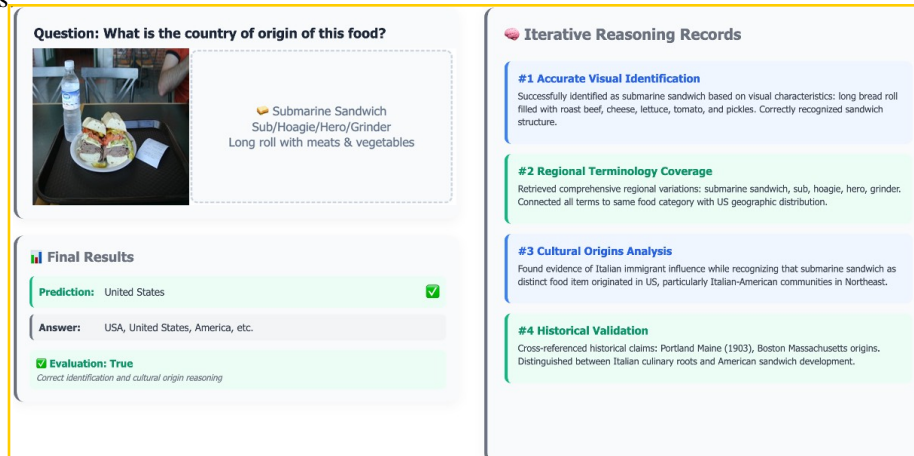


Figure A1: An initial correct visual entity leads to the retrieval of its corresponding knowledge.

Compositional Reasoning with Heterogeneous Knowledge. MI-RAG demonstrates compositional reasoning by integrating visual grounding with textual knowledge. As shown in the Amrum Lighthouse example (Figure A2), it first performs a visual analysis, identifying a lighthouse on an elevated landform. Through iterative retrieval, it grounds the specific entity as the “Amrum Lighthouse” and retrieves textual evidence from the knowledge base stating the structure is built “atop a 25 metres high dune”. The final compositional reasoning step involves synthesizing the visual evidence of an “elevated landform” with this specific textual fact to correctly identify the feature as a dune.

Progressive Reasoning from Generic to Specific. This case demonstrates the model’s ability to progressively refine its understanding through iterative refinement. When asked for the creator of the object in Figure A3, the model’s initial reasoning record correctly notes that a specific creator

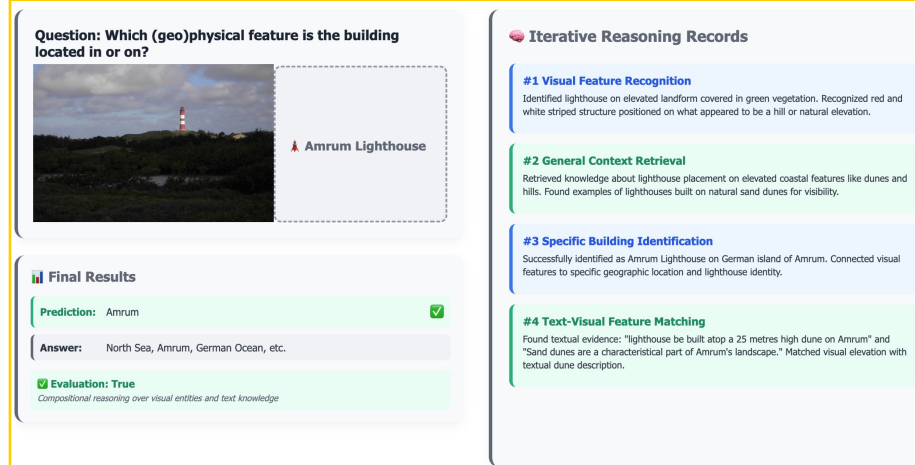


Figure A2: A visual entity is linked with corresponding textual knowledge.

of the telescope cannot be determined from the image alone. However, it successfully classifies the object as a Newtonian telescope in subsequent iterations. This crucial intermediate step enables a subsequent query generation focused on the inventor. This refined query retrieves the correct historical fact, showcasing how MI-RAG leverages reasoning records to guide the progression from a generic classification to a specific answer.

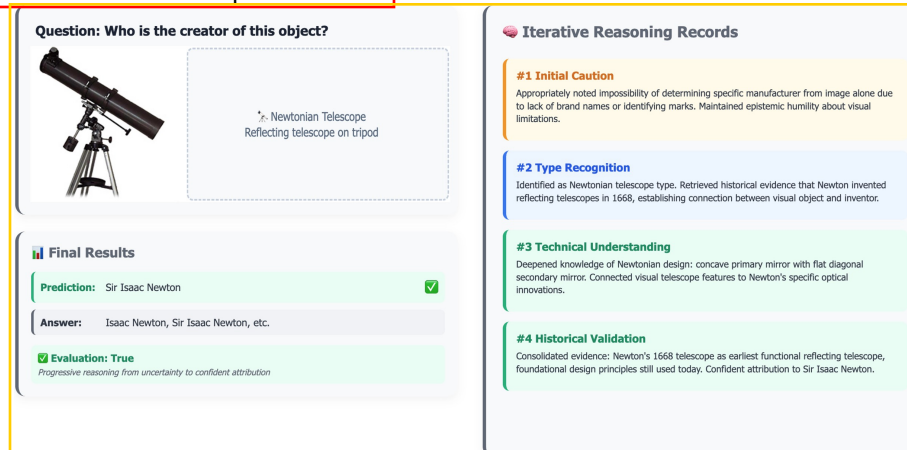


Figure A3: Progressively refine its understanding from generic object recognition (Telescope) to specific knowledge of an entity (Newtonian telescope, which was invented by Newton).

Compose Multi-Faceted Knowledge. This case demonstrates how MI-RAG answers questions that require linking multiple facets of knowledge. In Figure A4, it identifies architectural details (Russian Orthodox), the specific building's identity (Vysokopetrovsky Monastery), and historical facts about its founder through iterations. MI-RAG then performs compositional reasoning over this diverse evidence, correctly synthesizing the facts to conclude that the monastery is dedicated to Saint Peter of Moscow.

A.2 FAILURE CASE ANALYSIS

To better understand the limitations of our model, we present a qualitative analysis of representative failure cases. Each case provides insights into areas for future improvement by pairing the visual example directly with its analysis.

Intent Misinterpretation. A primary failure mode involves misunderstanding the user's query, even with correct evidence. In Figure A5, the model is asked for the "closest parent taxonomy" of

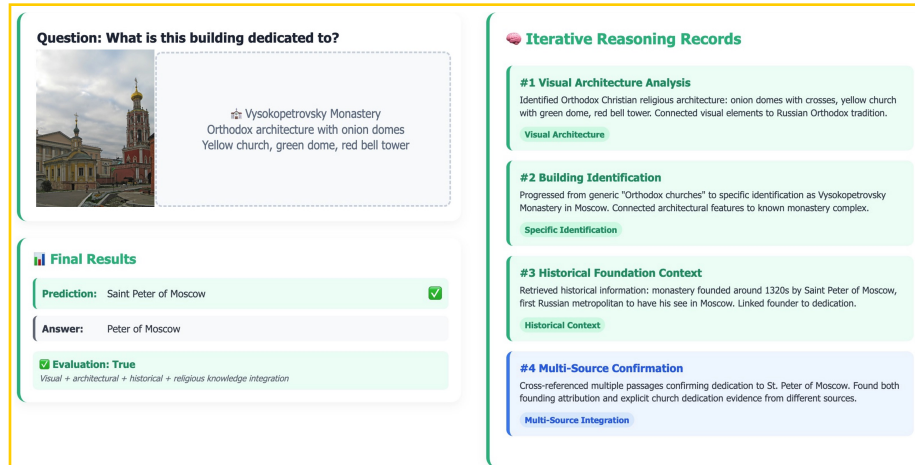


Figure A4: Multi-Faceted Knowledge Synthesis: Diverse aspects of knowledge, including architectural and historical facts, are retrieved and synthesized to answer.

a peccary. Despite correctly identifying the animal and its full taxonomic hierarchy, it misinterprets the query's intent, providing the suborder (Suina) rather than the more immediate genus (Pecari).

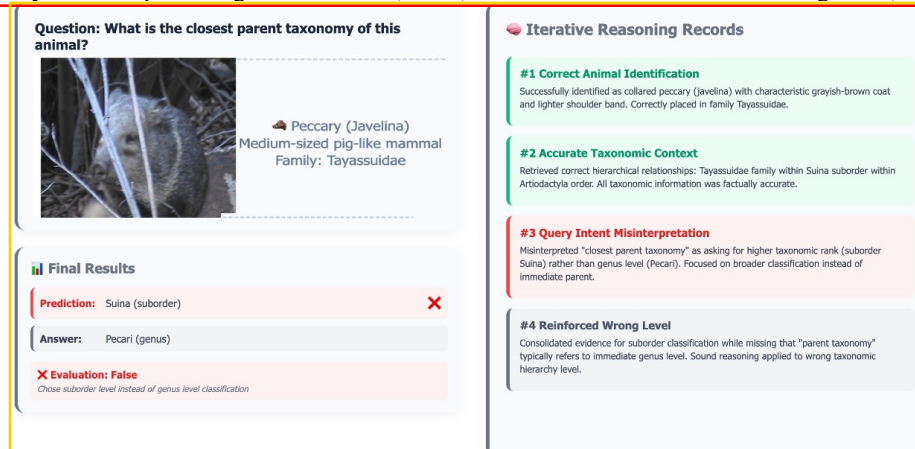


Figure A5: Intent Misinterpretation: The model misunderstands the required taxonomic specificity.

Entity Misidentification. Failures can also arise from the initial visual analysis stage, such as incorrectly identifying the primary subject. In Figure A6, the model correctly identifies the general scene as a Swiss lake, but incorrectly identifies the specific body of water as Lake Walen instead of Lake Lucerne. This initial error leads to a wrong subsequent retrieval and concludes with wrong answer.

Subject Selection Error. Another form of visual grounding error occurs when the model focuses on a non-primary subject in the scene. Figure A7 shows a scene with two aircraft. The model correctly identifies both but erroneously selects the second aircraft as the subject of the query, derailing the entire subsequent reasoning process.

Incomplete Knowledge Retrieval. This class of errors arises from limitations in the knowledge base or the retrieval process. In Figure A8, the model identifies a bird as a New World Oriole but fails to retrieve knowledge on the specific genus (Pitangus). Its final answer is therefore based on incomplete evidence, leading to an incorrect conclusion.

Insufficient Visual Evidence. Sometimes, a query is unanswerable from the image, since it provides generic entities. The image in Figure A9 shows a generic classroom, which lacks sufficient

Question: Where is the lake outflow to?



📍 Swiss Lake
Located in Switzerland
Actual: Lake Lucerne (Eugenisee area)

Final Results

Prediction: Lake Zurich via Linth River ❌

Answer: Reuss River

❌ Evaluation: False
Correct reasoning about wrong lake (Walensee vs Eugenisee)

Iterative Reasoning Records

#1 Correct Initial Assessment
Correctly identified as Eugenisee in Engelberg, Switzerland. Appropriately noted that outflow information was not visible in image or provided text about Eugenisee.

#2 Lake Entity Confusion
Suddenly shifted from Eugenisee to discussing Lake Engstlensee, then to Walensee. Lost track of the original lake identity despite correct initial identification.

#3 Accurate Walensee Research
Retrieved comprehensive and accurate information about Walensee: fed by Seer, Murgbach, Linth rivers; outflows to Lake Zurich via Linth. All geographic facts were correct.

#4 Confident Wrong Conclusion
Consolidated evidence about Walensee's outflow to Lake Zurich. Reasoning was geographically sound but applied to completely wrong lake entity from the original image.

Figure A6: Entity Misidentification: An initial, incorrect identification of the primary subject.

Question: What country does this organization belong to?



✈️ Two Aircraft Scene
Foreground: OM-USS
Background: Austrian arrows (Austria)

Final Results

Prediction: Austria ❌

Answer: United Kingdom, UK, Britain, etc.

❌ Evaluation: False
Focused on background aircraft instead of primary foreground subject

Iterative Reasoning Records

#1 Multi-Aircraft Recognition
Correctly identified two aircraft: foreground business jet (OM-USS) and background Austrian arrows aircraft (OE-LVP). Recognized scene complexity with multiple potential subjects.

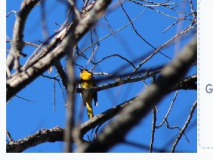
#2 Background Aircraft Focus
Chose to focus on Austrian arrows aircraft in background rather than prominent foreground business jet. This subject selection determined the entire reasoning direction.

#3 Accurate Austrian Airlines Research
Retrieved comprehensive information about Austrian Airlines and Tyrolean Airways. Correctly connected Austrian arrows branding to Austria and OE registration prefix.

#4 Confident Wrong Subject
Consolidated evidence about Austrian airlines' nationality. All reasoning was accurate for chosen aircraft but missed that primary subject should have been the foreground aircraft.

Figure A7: Subject Selection Error: The model focuses on a background subject instead of the primary subject.

Question: What is the closest upper taxonomy of this bird?



🐦 New World Oriole
Yellow body, black head/wings
Genus: Icterus (retrieved), Pitangus (missed)

Final Results

Prediction: Icteridae (family) ❌

Answer: Pitangus (genus)

❌ Evaluation: False
Retrieved Icterus genus but missed Pitangus genus information

Iterative Reasoning Records

#1 Correct Visual Identification
Successfully identified bird with vibrant yellow body and black head/wings. Accurately described plumage coloration and body shape characteristics from visual analysis.

#2 Incomplete Genus Coverage
Retrieved information about Icterus genus and New World orioles but failed to search for or retrieve information about Pitangus genus. Knowledge base search had insufficient coverage.

#3 Accurate Icteridae Research
Retrieved comprehensive information about Icteridae family (New World blackbirds). All taxonomic details about family classification and relationships were factually correct.

#4 Limited by Retrieval Gap
Concluded with Icteridae family based on available evidence. Reasoning was sound but constrained by incomplete retrieval that never discovered the actual Pitangus genus.

Figure A8: Incomplete Knowledge Retrieval: Fails to retrieve knowledge about the correct entity.

visual features to determine a specific location. The model correctly identifies the generic context but fails to recognize the query's unanswerable nature, producing a non-specific answer.

Question: What is the location of this facility?



Generic Classroom
Whiteboard with equations
No identifying features visible

Final Results

Prediction: Classroom (generic) ❌

Answer: Regenstein Library

❌ Evaluation: False
Image lacks visual markers for specific facility identification

Iterative Reasoning Records

#1 Correct Initial Assessment
Initially recognized that location cannot be determined from image: "no identifiable landmarks, signs, or unique architectural features." Correctly noted lack of specific visual markers.

#2 Shifted to Generic Classification
Despite initial correct assessment, began reasoning about generic "classroom" or "educational institution" rather than maintaining that specific location was indeterminable.

#3 Generic Context Reasoning
Retrieved general information about whiteboards in educational settings. Applied broad contextual knowledge rather than acknowledging visual evidence limitations.

#4 Generic Conclusion
Concluded with "classroom" answer when specific facility identification was required. Failed to maintain that question was non-answerable from available visual information.

Figure A9: Insufficient Visual Evidence: The query is unanswerable from the image provided.

Reasoning-Answer Discrepancy. Finally, some failures occur when the reasoning is sufficient, but the final answer does not align with the expected format. In Figure A10, the model correctly deduces that the Hawker 800 is a development of the British Aerospace 125. However, its prediction is marked as incorrect because the ground truth expects a different but related entity name, highlighting a format mismatch issue rather than a fundamental reasoning error.

Question: What is the immediately prior item that this organization follows in a series?



Hawker 800 Aircraft
Registration: XA-GTE
From AIRLINERS.NET database

Final Results

Prediction: British Aerospace 125 ❌

Answer: Hawker Siddeley, BAC, Siemens Plessey, etc.

❌ Evaluation: False
Sound reasoning but format mismatch with expected answers

Iterative Reasoning Records

#1 Initial Misinterpretation
The model initially focused on "AIRLINERS.NET" copyright text visible in the image, incorrectly assuming the question referred to a website naming series rather than the aircraft itself.

#2 Aircraft Recognition
Successfully identified the aircraft as a Hawker 800 (registration XA-GTE) but lacked sufficient contextual information about its developmental lineage to answer the series question.

#3 Context Retrieved
Retrieved critical developmental context: "British Aerospace 125... Later variants marketed as Hawker 800." Established chronological series: British Aerospace 125 → Hawker 800.

#4 Evidence Consolidated
Cross-referenced multiple passages confirming developmental relationship. Found consistent evidence: "Hawker 800 is a development of British Aerospace BAe 125" but answer format differed from expected alternatives.

Figure A10: Format Mismatch: The model's reasoning is sufficient, but the output format differs from the expected answer.

B THE USE OF LARGE LANGUAGE MODELS

In accordance with ICLR 2026 policy, we disclose the use of a large language model (LLM) as an assistive tool in the preparation of this manuscript. The LLM's role was confined to that of a writing and formatting assistant. Specific tasks included improving grammar, clarity, and style, as well as providing support for formatting LaTeX code, such as adjusting the layout and style of figures and captions. All conceptual contributions, research ideation, experimental design, data analysis, and the final written conclusions were generated exclusively by the human authors. The authors take full responsibility for all content presented in this paper, including its scientific accuracy and integrity.

C PROMPT TEMPLATES FOR MI-RAG

```

1  # 1. Initial description generation
2  prompt = f"Question: {question}\n Concisely describe image which is
   ↳ relevant to question.\n"
3
4  -----
5  # 2. Query Expansion
6  prompt = f"Question: {question}\n{reasoning_record}\n"
7
8  -----
9  # 3. Query Generation Prompt
10 Question: {question}
11 Knowledge: {reasoning_records}
12
13 Please first analyze all the information in a section named Analysis
14 ↳ (## Analysis).
15 Generate two follow-up questions to search for additional information
16 ↳ and helpful to confirm knowledge, in a section named Queries (##
17 ↳ Queries).
18 Your output should be in the following format:
19
20 ## Analysis
21 Analysis question and knowledge to ask context-specific queries that
22 ↳ helps to address question.
23 ## Queries
24 Question 1: question 1.
25 Question 2: question 2.
26
27 -----
28 # 4. Reasoning Record Generation Prompt
29 # This prompt instructs the model to synthesize information into a
30 ↳ coherent summary.
31
32 Question: {question}
33 Knowledge: {knowledge}
34
35 Based on image, description and knowledge, summarize correct and
36 ↳ relevant information with image and question.
37
38 -----
39 # 5. Final answer prompt
40 Please answer the following question using the provided information and
41 ↳ image.
42
43 Question: {question}
44 Relevant Knowledge: {reasoning_records}
45
46 Based on the information, provide a detailed answer to the question.

```

Figure A11: Pseudo-code for prompt templates used in our MI-RAG framework. The Query Generation prompt drives the iterative retrieval process, while the Reasoning Record Generation prompt synthesizes information at each step.

Our Multimodal Iterative RAG (MI-RAG) framework employs structured prompts to dynamically refine its understanding and gather relevant knowledge. To generate initial reasoning records, we use a query-focused description of the input image to retrieve an initial set of knowledge. The core iterative loop then utilizes two prompts. First, the query generation component of multi-query uses the prompt to instruct the model to analyze the accumulated reasoning records and generate two questions. This step is crucial for progressively searching for more specific and helpful information. For the step expansion component of multi-query, we simply concatenate the query with the previous reasoning record. Second, the reasoning record generation prompt directs the model to synthesize the image and newly retrieved knowledge into a query-focused summarization. This rea-

soning record is the result of compositional reasoning and improves subsequent retrieval iterations, and is used to derive the final answer.

D MULTIMODAL RETRIEVAL IMPLEMENTATIONS

To build our knowledge base, we processed the Wikipedia dataset from the Encyclopedic VQA benchmarks, which contains 2 million image-section pairs. The construction process involves generating a multimodal embedding for each pair and indexing them for efficient retrieval. For each sample, we extracted a normalized image embedding using a pretrained SigLIP model and a normalized text embedding using a pretrained ModernBert model, which is trained with the GTE method. These two vectors are then concatenated to form a multimodal embedding. Finally, all embeddings are indexed using FAISS with an IndexFlatIP (Inner Product) to create a knowledge base. A corresponding metadata file was also created to map the index entries back to their original image paths and Wikipedia text sections. When encoding the text embedding, we use a summary of each Wikipedia section. For the large retriever configuration of SigLIP2-g and Qwen3-Embedding-0.6B, we employ the image-to-mixture similarity proposed in RA-CM3 Yasunaga et al. (2023) and encode mixed embedding instead of image embedding.

At query time, we encode the input image and refined text query into the same multimodal embedding space as our knowledge base. We first generate normalized embeddings for the image and the text query separately. These vectors are then concatenated to form a single multimodal query vector. Finally, we perform a maximum inner product search against the knowledge base to retrieve the top k entries with the highest multimodal similarity.

Table A1: Performance analysis of multimodal similarity on the InfoSeek validation split and Encyclopedia test split.

Retriever	Dataset	Query Modality	R@1	R@5	R@10
SigLIP2-SO400m+ ModernBert-GTE	InfoSeek	image	52.5	60.2	68.3
		image+text	57.8	65.3	72.3
	Encyclopedia	image	30.8	36.6	41.8
		image+text	34.9	40.5	45.3
SigLIP2-g+ Qwen3-Embedding-0.6B	InfoSeek	image	66.7	72.7	77.5
		image+text	69.4	76.2	81.1
	Encyclopedia	image	36.2	41.9	46.4
		image+text	43.1	48.7	54.5

As shown in Table A1, we evaluated the retriever performance on the InfoSeek validation and Encyclopedia test splits. The results confirm two key points. First, using a multimodal query (image+text) that considers both visual and textual similarity consistently yields higher recall than using a unimodal (image) query alone. Second, we observed that our scaled retriever configuration (SigLIP2-g + Qwen3-Embedding-0.6B) performs better than the more lightweight setup (SigLIP2-SO400m + ModernBert-GTE). Despite the performance gap, we used the lightweight setup for our main experiments. This choice ensures a fair comparison and emphasizes that our framework’s primary improvements stem from its iterative reasoning process, not from relying on a stronger base retriever.

E FEW-SHOT PROMPTING FOR ANSWER EXTRACTION

To evaluate performance on the Exact Match metric for the InfoSeek dataset, we employed a few-shot prompting strategy to guide the model toward generating an exact answer. For each question in the validation set, we provided the model with three demonstrations drawn from the training split, selected based on question similarity. These demonstrations consist of an image, a context of reasoning records, a question, and the ground-truth answer. By presenting these structured examples, we condition the model to generate responses in a specific format, which minimizes extraneous text and focuses on producing the exact entity or fact required by the question. This in-context learning approach helps to align the model’s output with the strict requirements of the EM metric.

```

1 Answer the knowledge-intensive question based on the provided image and
  ↳ context.
2 Generate a concise and accurate answer grounded in the retrieved
  ↳ information.
3 Use the context to support reasoning, and directly output the final
  ↳ answer.
4
5 Three examples are shown below:
6
7 ##Example 1:
8 [Image 1 Content]
9 ##Context: {few_shot_context_1}
10 ##Question: {few_shot_question_1}
11 ##Best Answer: {few_shot_answer_1}
12
13 ##Example 2:
14 [Image 2 Content]
15 ##Context: {few_shot_context_2}
16 ##Question: {few_shot_question_2}
17 ##Best Answer: {few_shot_answer_2}
18
19 ##Example 3:
20 [Image 3 Content]
21 ##Context: {few_shot_context_3}
22 ##Question: {few_shot_question_3}
23 ##Best Answer: {few_shot_answer_3}
24
25 Now, answer this question
26 [Main Image Content]
27 ##Context: {reasoning_records}
28 ##Question: {question}
29 ##Best Answer:

```

Figure A12: The few-shot prompt template used to guide the model for the Exact Match evaluation.

F COMPUTATIONAL COST ANALYSIS

Table A2: Computational cost per sample for each iteration of the MIRAG framework

Iteration	API Time (s)	Retrieval Time (s)
Iteration 0	7.96	15.23
Iteration 1	19.89	46.36
Iteration 2	32.29	79.74
Iteration 3	41.71	105.384

We analyze the per-sample computational cost of MI-RAG on a commodity server (NVIDIA RTX 3090, Intel Xeon Gold 6248R), performing retrieval locally and using the OpenRouter API OpenRouter (2025) for MLLM inference. As detailed in Table A2, both API and retrieval times scale linearly with each iteration, an expected characteristic of our design. Specifically, API time may increase as accumulated reasoning records lengthen the context for query generation, while retrieval time grows due to the increased processing required to encode these context-enriched queries. Crucially, this latency is not a fundamental limitation of our framework. The overhead stems primarily from the serialized execution of queries using vanilla index configurations and network overhead from successive API calls with multiple images and texts.

G SUBSET DOWNSAMPLING PROCEDURES

To support an efficient analysis and design of our model, MI-RAG, we perform ablation studies on a compact and diverse subset of each dataset. We generate this subset using the iterative downsampling procedure detailed in Figure A13, which aims to eliminate redundancy.

```

1  # D: Input Dataset of questions
2  # Q: A single question from the dataset
3  # Emb_Q: The text embedding of a question Q
4  # t: The similarity threshold
5  # subset: Downsampled dataset
6
7  function DownSampleDataset(D, t):
8      subset = []
9      subset_emb = []
10
11     for Q in D:
12         is_redundant = False
13         Emb_Q = Encode(Q)
14
15         for emb in subset_emb:
16             if Similarity(Emb_Q, emb) > t:
17                 is_redundant = True
18                 break
19
20         if not is_redundant:
21             subset.append(Q)
22             subset_emb.append(Emb_Q)
23
24     return subset

```

Figure A13: Pseudo-code for downsampling: discard questions whose similarity to any in the subset exceeds threshold, yielding a non-redundant subset.

The algorithm greedily constructs a subset by adding a new question Q from the original dataset D only if it is sufficiently dissimilar to all questions already selected. Specifically, a question is considered redundant and discarded if the similarity of its embedding, Emb_Q , exceeds a predefined threshold with any embedding already in the subset. This threshold is adjusted for each dataset to yield a final subset containing approximately 500 diverse question-answer pairs.

H QUERY STRATEGY ANALYSIS

Table A3: Analysis of each query formulation’s contributions to retrieval recall from each KB on the InfoSeek validation subset.

Method	Multimodal KB Recall (%)	Textual KB Recall (%)	Hetero. KB Recall (%)
Query Expansion	90.02	87.44	93.08
Query Generation	88.24	80.03	90.80
Multi-Query	90.50	90.02	94.36

In this section, we analyze the contributions of each query strategy to retrieval performance on the heterogeneous KB. Query expansion proves highly effective at retrieving directly related information, achieving 93.08% recall. In contrast, query generation is designed to explore complementary reasoning paths. Because its retrieval budget is divided between two generated sub-queries, its recall is slightly lower than that of the query expansion. However, its primary value lies in finding a diverse set of factual links that a direct expansion might overlook.

I QUERY TRANSFORMATION VS. REASONING

To assess which component of our framework benefits from increased model capacity, we performed an analysis by switching a lightweight model (Gemma-3-4B) and a large model (Gemini-2.5-Flash) between query transformation and reasoning generation. As shown in Table A4, allocating a more capable model to query transformation yields higher recall, likely due to improved query precision

Table A4: Impact of allocating a larger model (Gemini-2.5-Flash) to either the query transformation or reasoning records generation, while a lighter model (Gemma-3-4B) handles the other. Evaluated on the InfoSeek validation subset.

Query Transformation	Reasoning Generation	Acc (%)	Recall (%)
Gemma-3-4B	Gemini-2.5-Flash	56.36	91.63
Gemini-2.5-Flash	Gemma-3-4B	54.59	93.56

and diversity. However, assigning the large model to reasoning records generation results in higher overall accuracy, indicating that this configuration provides a benefit to the final accuracy.